

Cole Thomas Fenner

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Professional Summary

Mechanical Engineer and Robotics CAD Designer with hands-on experience designing mechanical systems from fixtures to motorized gearbox assemblies. spanning concept through a CNC-manufactured prototype. Research background in kinematic design, motor controller integration, and ROS.

Education

Oregon State University— Corvallis, OR

M.S. in Robotics (GPA: 4.0) | Graduated: Sept 2024 – May 2025

Research Member, Laboratory for Robotics and Applied Mechanics (LRAM)

- ❖ Thesis: Projectile arm thrower that isolated angular momentum. Use of Odrive controllers and ROS.
- ❖ Focus: Adjoint matrices in group-based design, compliant Kinova arm control, optimized path planning, composite mechanics.

B.S. in Mechanical Engineering (GPA: 3.9) | Graduated: Sept 2021 – Aug 2024

- ❖ Focus: Composite Manufacturing, 3D Modeling, Design, GD&T
- ❖ Awards: OSU Finley Scholarship, Intel Scholarship, Dean's List, Honor Roll

Technical Skills

- ❖ **Programming & Controls:** Python, ROS, Matlab, Gazebo, HTML, C++, CAN bus
- ❖ **3D Modeling:** Siemens NX, SolidWorks, Fusion 360, Autodesk Inventor, AutoCAD, Onshape
- ❖ **Prototyping:** CNC machining, 3D printing, laser cutting, welding, sheet metal bending, pneumatics
- ❖ **Engineering:** Stress simulations, GD&T, composite manufacturing, lean Six Sigma, adjoint matrix coding

Engineering Experience

Mechanical Engineer with Meta via Sigma Design / Aug 2026 – Present / Redmond, WA

- ❖ Design and fabricate fixtures to support prototype assembly and quality control of actuators.
- ❖ Use NX and SolidWorks to support designers on reworks and design of supporting components.
- ❖ Integrate and program prototype motor controllers for automation of test fixtures.

Engineering Technician II with Meta via Sigma Design / Feb 2026 – Aug 2026 / Redmond, WA

- ❖ Contributed to Meta hardware development through hands-on assembly, testing, and documentation of actuators, gearboxes, and mechanical systems.
- ❖ Assisted mechanical engineers by providing practical design feedback during prototyping.

First Robotics Competition Mentor (Volunteer), Shockwave 4488 / Sept 2025 – Feb 2026

- ❖ Mentor high school students in mechanical design, CAD modeling, and manufacturing using Onshape.

Graduate Teaching Assistant, Oregon State University / Sept 2024 – June 2025

- ❖ Assisted with engineering coursework by grading, proctoring exams, and holding student office hours.

Design Engineering Intern, MEGI Engineering Inc. / June 2022 – Sept 2022

- ❖ Created detailed 3D models of industrial paper mill systems using point cloud scan data.
- ❖ Updated piping and electrical systems in AutoCAD to improve infrastructure adaptability. (P&ID)

First Robotics Competition Team Captain, Shockwave 4488 / Aug 2017 – Mar 2021

- ❖ Designed, prototyped, and manufactured competition robots using Fusion 360 and CNC machining.
- ❖ 30" x 30" robots fabricated from aluminum and polycarbonate by CNC plates, bars, and sheet metal.
- ❖ Led a 30+ team to achieve 10th in the world recognition, including:
 - 2018 World Galileo Division Champion, Innovation in Control Award.
 - 2019 Pacific Northwest Championship finalist, Industrial Design Award.
 - 2020 District Event Winner, Industrial Design Award.
- ❖ Led outreach initiatives, adapting toys for children in China with disabilities to improve accessibility.

Projects

Robotics Master's Project, Oregon State University / Sept 2024 – June 2025

- ❖ Manufactured a 2-DOF decoupled spin projectile launcher using a virtual 4-bar belt drive.
- ❖ Implemented synchronous motion control with ODrive motor controllers and ROS2 CAN.
- ❖ Integrated an ESP32 microcontroller to remotely trigger pneumatic mechanisms.

SAE Baja Senior Project, Oregon State University / Sept 2023 – March 2024

- ❖ Led reverse engineering and CNC manufacturing of a limited-slip differential gearbox.
- ❖ Designed precisely toleranced, interchangeable disc springs with controlled breakaway torque.
- ❖ Conducted FEA simulations to optimize stress tolerances and enhance durability.
- ❖ Collaborated with a multidisciplinary team to integrate the gearbox into the SAE Baja vehicle.

Swerve Drive System, 4488 Shockwave / 2020

- ❖ Designed and CNC-fabricated a custom corner packaging differential gearbox for a swerve module.
- ❖ Integrated custom subsystems including intakes, funnels, arm manipulators, and shooter mechanisms.

Honors & Activities

Tennis Officer, OSU Club Team / 4 years

Chess Founder & Competitor, 2nd Place Individual State, Multi-time Regional Champion | 10 years

Tennis Team Captain, Glencoe HS Varsity / 4 years

Tae Kwon Do, Black Belt and Mentor / 5 years